

TryNex Lifestyle

Full Technical Documentation

Premium Custom Apparel · Bangladesh

Platform:	Full-stack e-commerce + Design Studio
Market:	Bangladesh (all 64 districts)
Domain:	trynexshop.com
API:	trynex-api.onrender.com
Stack:	React 19 + Vite 7 + Express 5 + PostgreSQL
Generated:	29 May 2026

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§1 Platform Overview & Architecture

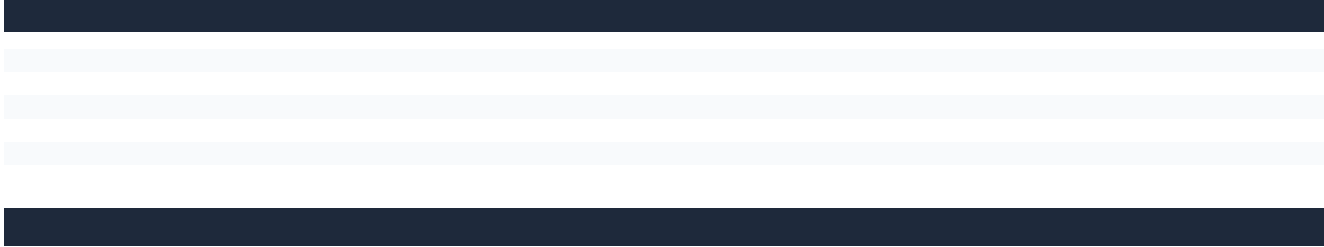
Dev Environment	Replit (NixOS)	Replit	Both services run side-by-side, pnpm ws
Cache	Upstash Redis (REST)	Upstash	Settings, product list, rate limit counters
Full-stack e-commerce platform			Product images, original design files
			Monorepo Layout
			• artifacts/trynex-storefront/ — React 19 + Vite
			• artifacts/api-server/ — Express 5 REST A
			• artifacts/api-worker/ — Cloudflare Worker
			• artifacts/mockup-sandbox/ — Isolated Vite c
			• lib/db/ — Drizzle schema, 12 mi
			• lib/api-spec/ — OpenAPI 3 spec + g
			• lib/api-zod/ — Generated Zod valida
			• lib/api-client-react/ — TanStack Query ho
			• scripts/ — Seed, PDF generation
			• .github/workflows/ — deploy.yml (main

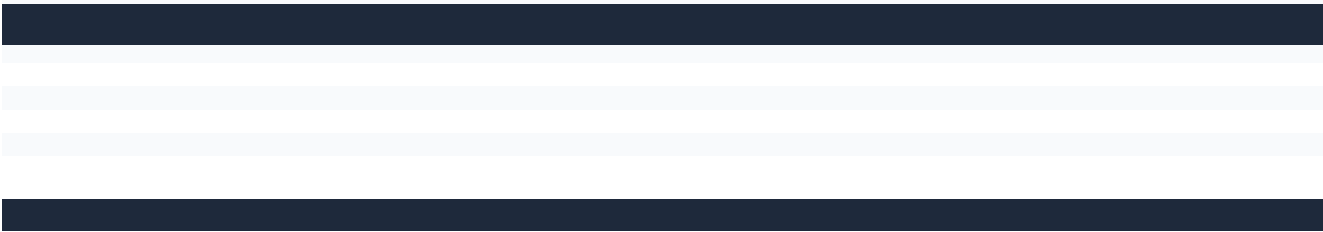
@upstash/redis
DOMPurify

latest
latest
latest

- In-app Redis REST client
- Sanitise user-generated HTML (blog content)
- Non-blocking toast notifications
- Client-side order summary / PDF generation

- Cloudflare R2 — object storage for product images + design originals
- Cloudflare Pages — static SPA hosting, global CDN, custom domain t
- Render.com — Node 24 web service for Express API (auto-deploy on
- Upstash Redis — REST-based Redis for caching and rate-limit counte
- Neon PostgreSQL — serverless PG fallback chain (DATABASE_URL_
_FAILOVER)
- GitHub Actions — 5-job CI/CD: TypeCheck ! CF Pages ! CF Worker !





§5 Database Schema (19 Tables)

products	id, name, slug, price, discountPrice, categoryId, createdAt, updatedAt, deletedAt, viewCount, index	Product catalogues
categories	id, name, slug, createdAt, updatedAt, deletedAt, viewCount, index	Product categories
orders	id, orderNumber(TY-XXXXX), customerName, customerEmail, customerPhone, shippingAddress, shippingDistrict, paymentMethod, paymentStatus, status, items[], subtotal, shippingCost, total, promoCode	All orders
blog_posts	id, title, slug, content, excerpt, featuredImage, category, tags[], published, publishedAt, viewCount, seoTitle, seoDescription	Blog articles
customers	id, email(unique), phone, passwordHash, name, address, googleId, facebookId, referralCode(unique), referredByCode, totalOrders, totalSpent, utmSource, utmMedium, utmCampaign, discountType(ratio/percent), approved, createdAt	Registered customers
testimonials	id, customerName, message, rating, productId, discountValue, minOrderValue, maxUses, usedCount, expiresAt, active, isVirtual	Customer testimonials
promo_codes	id, code, discountValue, minOrderValue, maxUses, usedCount, expiresAt, active, isVirtual	Discount codes
reviews	id, productId, description, price, rating, createdAt, updatedAt, deletedAt, viewCount, index	Product reviews
packages	id, name, slug, description, price, status, createdAt, updatedAt, deletedAt, viewCount, index	Product packages
region_stats_logs	id, regionId, createdAt, updatedAt, deletedAt, viewCount, index	Region statistics tracking
newsletter_subscribers	id, email(unique), createdAt, updatedAt, deletedAt, viewCount, index	Newsletter subscribers
order_messages	id, orderId, senderId, receiverId, message, status, createdAt, updatedAt, deletedAt, viewCount, index	Order message threads
undo_transactions	id, transactionId, rollbackData, createdAt, updatedAt, deletedAt, viewCount, index	Undo transactions (rollback data)

§6 Authentication System

Customer Auth

- Registration: email + password (bcrypt 10 rounds) + phone (required for COD)
- Login returns: short-lived JWT access token (1h) + HttpOnly refresh cookie (30d)
- Refresh: POST /api/auth/refresh reads HttpOnly cookie, issues new access token
- Google OAuth 2.0 via /api/auth/google callback (GOOGLE_CLIENT_ID required)
- Facebook OAuth via /api/auth/facebook callback (FACEBOOK_APP_ID required)
- Guest checkout: order placed without account !' auto-creates customer record !' emails credentials
- Password reset: email token (1h TTL) !' POST /api/auth/reset-password
- JWT_SECRET must be set; defaults to random on dev (stateless, not revocable)

Admin Auth

- Single admin account seeded via ADMIN_PASSWORD at startup (bcrypt 10 rounds)
- Login returns: 30-min admin JWT signed with ADMIN_JWT_SECRET (separate from customer secret)
- Sessions stored in admin_sessions table — each token hash is persisted and revocable
- Optional TOTP 2FA: enable in admin !' generates otplib TOTP secret !' scan QR in authenticator app
- All admin routes guarded by verifyAdminToken middleware (validates JWT + checks not-revoked in DB)
- Admin JWT secret must be 32+ chars in production (hard exit at boot if too short)
- Admin sessions expire after 30min of inactivity (lastUsedAt rolling window)
- IP address logged per session for audit purposes

Security Headers

- helmet() sets: X-Content-Type-Options, X-Frame-Options (SAMEORIGIN), X-XSS-Protection, HSTS (prod)
- CORS: strict ALLOWED_ORIGINS allowlist; production refuses to start if ALLOWED_ORIGINS is unset
- Content-Security-Policy: configured via helmet; frames blocked except SAMEORIGIN
- No credentials are ever logged; Pino logging masks passwords/tokens by field name

§7 Design Studio — Technical Deep Dive

The Design Studio (/design-studio) is the most complex feature. It renders a 1000×1000 SVG canvas with real-time photographic garment mockups, a drag-and-drop layer system, AI-powered tools, and a 3D preview. It is self-contained in artifacts/trynex-storefront/src/pages/DesignStudio.tsx (~4300 lines).

Canvas & Coordinate System

- All products share a unified 1000×1000 SVG viewBox
- PrintZone {x, y, w, h} defines the printable area per product face in 1000-unit space
- Layer positions are stored in the same 1000-unit space — compositing is coordinate-space-invariant
- Canvas renders inside a <svg> element; layers are absolutely-positioned children
- Pinch-to-zoom and pan are implemented via pointer events on the SVG container
- Undo/redo stack: full layer state snapshot on every mutating action (30-deep ring buffer)

Photographic Mockup System (GarmentSVG)

- All garments use real studio photography (transparent PNG, 1000×1000 px)
- Colour rendering: isNearBlack(hex) = luminance < 0.12 !' use dedicated black photo; all other colours use white base photo + SVG multiply-tint filter
- SVG tint filter pipeline: desaturate !' feFlood(tintColour) !' feComposite(IN sourceAlpha) !' feBlend(multiply, grayscale) !' feComposite(IN sourceAlpha) = realistic fabric colour
- isLightTint(hex) = luminance > 0.92 !' no tint applied (white garment shows as-is)
- Curvature vignette: radial gradient darkens edges of garment photo to simulate 3D fabric wrap
- Highlight vignette: radial gradient at top-center adds subtle 3D volume effect

600px x:348 y:278 w:290 h:575
waterbottle
can
x:342 y:305 w:316 h:248

Layer System

- Layers: each has id, type (e.g. 'waterbottle', 'can', 'neck-label')
- Image layers: src (R2 URL)
- Text layers: text, fontFamily, size, color
- Multi-select: Shift+click; group layers
- Layer lock: locked layers cannot be moved
- Copy front/back helper: duplicate or move to front/back
- Design drafts: auto-save every 30s

AI & Enhancement Tools

- AI Art generation: POST /api/generate-art
- Background removal: POST /api/remove-bg
- HD Upscale: POST /api/upscale
- Custom colour picker: SASS color functions
- Template presets: 6 built-in (T-shirt, Hoodie, etc.)
- Fonts: 20 typefaces included

§8 3D Viewer & Composer

React Three Fiber 3D Preview

- 3D view toggled via the 2D/3D switch in the Design Studio header
- T-Shirt + Long Sleeve: real GLB mesh (models/tshirt.glb, models/longsleeve.glb) with UV-mapped canvas texture
- Hoodie: GLB mesh (models/hoodie.glb) with separate body + hood materials
- Cap: GLB mesh (models/cap.glb) with front panel UV mapping
- Mug: procedural Three.js CylinderGeometry with wrap texture (360° repeat)
- Water Bottle: procedural TubeGeometry with carabiner assembly (cap + body + clip geometries)
- OrbitControls: drag to rotate, scroll to zoom, auto-damping for smoothness
- Garment colour tinting: ShaderMaterial with uniform for colour value; multiply blend in shader
- Canvas texture: composed by composer.ts at up to 2048×2048px for crisp print quality

composer.ts — Canvas Texture Composition

- `composeGarmentMockup(layers, printZone, baseHeight, options) !' HTMLCanvasElement`
- Renders each layer (image/text) onto a 2D canvas in printZone coordinate space
- destination-in alpha-mask clips the composed artwork to the print zone outline
- Supports front + back face composition for multi-face products
- Text rendering: uses same font stack as SVG preview for WYSIWYG consistency
- Output fed to Three.js as `CanvasTexture` (auto-updates when layers change)
- Same composer used for: 3D preview, 2048×2048 order thumbnail (sent to admin), cart preview card

\$9 Order & Payment Flow

cancelled
delivered

Cancelled (before shipping)
Confirmed, received by customer
Out for delivery

delivered

Payment Methods

- Cash on Delivery (COD): 15% advance required !' customer provides transaction ID
- bKash, Nagad, Rocket, Upay: manual MFS (mobile financial services)
- Card: manual card payment link or in-person (offline)
- Payment status flow: pending !' submitted (customer pays)
- Admin verifies by checking MFS app logs and updating status

Shipping Calculation (Server-Side)

- Free shipping threshold: ₹3,000 by default (configurable)
- Base shipping fee: ₹300 + 10% of order value (capped at ₹1000)
- Promo codes can grant freeShipping flag (overrides threshold)
- Both fees are computed server-side — client subtotal + shipping = total
- District is captured but not used for tiered pricing (flat rate)
- Hamper items: server re-computes price from real price

Custom Design Order Flow

- User designs in Design Studio !' 2048px canvas thumbnail generated
- On 'Add to Cart': thumbnail uploaded to R2 as thumb_nano.jpg, metadata saved to design.json
- Cart stores R2 thumbnail URL + presigned design URL
- On order placement: R2 URLs included in order item metadata
- Admin can download original design file via GET /api/design/{id}/original (presigned URL)
- studioAssetsMissing flag set if R2 upload fails during cart creation

§10 Admin Panel — Full Feature List

Dashboard (AdminOrders)

- Auto-refreshes every 3 seconds via React Query polling
- BroadcastChannel cross-tab sync: order status changes propagate to other open admin tabs instantly
- Order cards show: TN-XXXXX number, customer name, phone, address, total, items summary, payment status badge
- Quick status workflow buttons: Confirm !' Ship !' Mark Delivered
- Quick payment verification: Verify button sets paymentStatus=verified
- Bulk actions: bulk mark as shipped, bulk cancel
- Export: download orders as CSV (date range, status filter)
- Design download: per-order 'Download Design' button !' generates fresh presigned R2 URL
- Order messages: in-line chat thread per order for customer communication

Product & Content Management

- Products: CRUD, multi-image upload (drag-to-reorder), R2 storage, color variants (name+hex+image per variant)
- Categories: create/rename/delete (reassign products before delete), category image
- Blog: full rich-text editor, SEO meta fields, tag management, AI content assist, schedule publish
- Testimonials: create/approve/delete; shown on homepage
- Reviews: approve/delete customer product reviews
- Hero banners: image + link + CTA text, drag to reorder, enable/disable per banner
- Hampers: build gift hamper packages with product picker, pricing, packaging images
- Promo codes: flat/percent discount, min order value, max uses, expiry, virtual (with specific items)

Settings & Configuration

- Studio T-Shirt Price, Studio Mug Price — controls checkout price for Design Studio orders
- Studio T-Shirt Colors, Studio Mug Colors — comma-separated hex list for Design Studio colour swatches
- Free Shipping Threshold, Shipping Cost — overrides defaults (Ÿ3 S Â 6100)
- Site Name, SEO Description, OG Image — global SEO meta defaults
- Google Analytics ID, Meta Pixel ID — injected into storefront HTML at runtime
- Announcement Bar messages — scrolling ticker text
- WhatsApp number — used in the floating WhatsApp button
- Social links (Facebook, Instagram, YouTube, TikTok)
- TOTP 2FA toggle for admin account
- AI Assistant: uses OpenAI-compatible API (configurable endpoint + model)

			429
Storage upload	15 min	40	429
Remove-hq / AI tools	15 min	15	429

Additional Security Controls

- DOMPurify: all user-generated HTML (blog content) sanitized
- Presigned URLs: 15-minute TTL for private design files
- Admin JWT: separate signing secret, stored hashed in env
- ALLOWED_ORIGINS: production server hard-exits on mismatch
- Admin JWT secret: server hard-exits on startup if < 32 chars
- Storage backend guard: server hard-exits if storage rejected
- bcrypt: 10 rounds for all passwords; timing-safe comparison
- SQL injection: Drizzle ORM uses parameterised queries
- CSRF: JWT in Authorization header (not cookie) for client apps
- Input validation: Zod schemas on all POST/PUT endpoints

§12 SEO, Structured Data & Performance

Structured Data (JSON-LD)

- Organization — brand name, logo, sameAs links (FB, IG, YouTube)
- WebSite + SearchAction — enables sitelinks search box in Google
- ClothingStore (LocalBusiness) — address (Dhaka), geo, openingHours, telephone
- Product — on /product/:id; price, availability, rating, offers
- BlogPosting — on /blog/:slug; datePublished, author, image, wordCount
- BreadcrumbList — on all deep pages
- FAQPage — on /faq (and any product page with FAQ accordion)

Technical SEO

- React Helmet Async: per-route <title>, <meta description>, canonical, OG, Twitter card
- Sitemap: GET /api/sitemap.xml — includes all products + blog posts with lastmod + image tags
- robots.txt: disallows /admin/, /checkout/, /cart/; allows everything else
- Bilingual keyword targeting: English + Bengali (তৈরীকৃত স্টোর থেকে সরাসরি পাওয়া যায়)
- Canonical URL: always https://trynexshop.com (prevents dev/staging duplicate content)
- Open Graph: image, title, type, url per route; og:locale = en_US with hreflang note

Performance

- Vite manual chunking: vendor-motion (framer-motion), vendor-query (@tanstack), vendor-radix (all Radix)
- React 19 concurrent mode: Suspense boundaries around 3D viewer and lazy-loaded routes
- Image optimisation: product images served from R2 with Cloudflare CDN caching
- Preload hints in index.html: garment mockup PNGs with fetchpriority=high for LCP
- DNS prefetch: Google Fonts, GTM, FB Connect
- PWA: service worker (Vite Plugin PWA) caches static assets, offline fallback page
- React Query: staleTime = 5 min for products, 10 min for settings; background refetch on focus
- Lazy loading: React.lazy() for all admin pages and heavy studio components (3D viewer, AI panel)

§13 Object Storage & File Handling

3 (lowest)
?

Replit Object Storage
AWS S3

No cloud keys set
AWS_ACCESS_KEY_ID set (no R2)
R2_ACCESS_KEY_ID set

Dev environment only
Alternative S3-compatible provider

- R2 bucket name from R2_BUCKET env var
- S3 endpoint: `https://{accountId}.r2.cloudflarestorage.com`
- Object keys follow pattern: `products/{id}/{filename}`
- Public CDN: if `R2_PUBLIC_BASE_URL` is set
- for public images)
- Private design files: always presigned (15-min)
- Boot guard: server hard-exits with error if NO
- sharp integration: thumbnails generated server

§14 Caching Strategy

Upstash Redis (Server-Side)

- Client: @upstash/redis REST client (HTTP, not Redis)
- In-process Map fallback: if `UPSTASH_REDIS_REST_URL` is not set
- Cached: site settings (60s TTL), product list
- Cache invalidated on every write via cache.
- Rate limit counters stored in Redis using slice

React Query (Client-Side)

- staleTime: products = 5 min, settings = 10 min
- Background refetch on window focus: enabled
- Optimistic updates: cart add/remove updates
- Query key structure: `['products', filters]`, `['products', filters, sort]`
- Suspense mode used for 3D viewer texture

§15 Email, Notifications & Integrations

Telegram Bot

- TELEGRAM_BOT_TOKEN + TELEGRAM_CHAT_ID env vars required
- Sends notifications on: new order placed, order status change, new review submitted
- Admin can test the bot via admin panel !' Settings !' Telegram test
- Webhook endpoint: POST /api/telegram-webhook for two-way messaging (future feature)

Facebook Integration

- Admin panel page: /admin/facebook-import
- Imports Facebook Page posts as draft blog posts or product listings
- Requires FACEBOOK_APP_ID and page access token configured in admin settings

Google / Social Auth

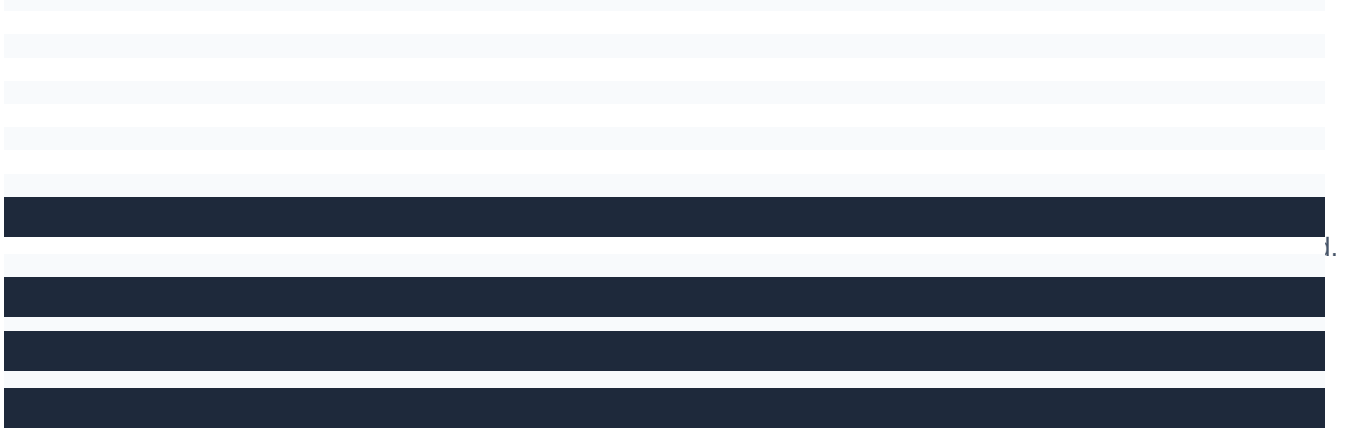
- Google OAuth: GOOGLE_CLIENT_ID required; callback at /api/auth/google
- Facebook OAuth: FACEBOOK_APP_ID required; callback at /api/auth/facebook
- Both create or link customer accounts on first login

AI Tools

- Background removal: fal.ai rembg API !' WASM browser fallback (no server key needed)
- HD Upscale: fal.ai 4× super-resolution (requires FAL_API_KEY)
- AI Art: POST /api/ai/generate !' configurable OpenAI-compatible endpoint
- Admin AI Assistant: streaming SSE chat; AI can preview and execute DB operations

§16 CI/CD & Deployment Architecture

deploy.yml	Push main / manual	TypeCheck: Upload Pages 2 with Worker	None
GitHub Actions	Pull Request to main	TypeCheck + Build (full build verification)	Worker, deploy jobs (continue-on-error)
Deploy Sequence (deploy.yml)			
• Job 0 — TypeCheck: pnpm install !			
• Job 1 — CF Pages (needs: typeche			
trynex-lifestyle-shop project			
• Job 2 — CF Worker (continue-on-er			
• Job 3 — Render (continue-on-error:			
not set)			
• Job 4 — Smoke Test (if: always): wa			
• Concurrency group: deploy-\$\$\$ githu			
ADMIN_PASSWORD	For Worker	Admin panel initial password	Required GitHub
ADMIN_JWT_SECRET	For Worker	Admin JWT signing secret (32+ chars)	
	For Worker	Customer JWT signing secret	



§18 GitHub Actions — Common Failures & Fixes

Smoke test fails on healthz
Render build state 00:00:00
Health check uses & h_not exit 1 — non-blocking
Updated: deploy skipped with warning if secret absent

Setting Up GitHub Secrets

- Go to: [https://github.com/\[owner\]/\[repo\]/settings/secrets](https://github.com/[owner]/[repo]/settings/secrets)
- Add New Repository Secret for each required variable.
- CLOUDFLARE_API_TOKEN: CF Dashboard !' My Profile Account:Read)
- CLOUDFLARE_ACCOUNT_ID: CF Dashboard !' bottom
- RENDER_DEPLOY_HOOK_URL: Render Dashboard !
- All other vars: copy from Replit Secrets panel
- Once set, re-run the last failed workflow: Actions !' Select

§19 Admin Onboarding Checklist

Initial Setup

- & Set ADMIN_PASSWORD in Replit Secrets (and Render env vars for production)
- & Set JWT_SECRET and ADMIN_JWT_SECRET (32+ chars) in Replit Secrets
- & Configure Cloudflare R2 bucket; set R2_* env vars
- & Set ALLOWED_ORIGINS to https://trynexshop.com in production
- & Set DATABASE_URL to production Neon or Render PG connection string
- & Add GitHub Secrets (see §18) to enable CI/CD
- & Verify custom domain: CF Dashboard !' DNS !' point trynexshop.com to CF Pages URL

First-Time Admin Setup

- & Login at /admin/login with ADMIN_PASSWORD
- & Enable TOTP 2FA (Admin !' Settings !' Security !' Enable 2FA) — scan QR in Google Authenticator
- & Set Studio Prices in admin Settings (T-Shirt Price, Mug Price, Bottle Price)
- & Set Studio Colours list (comma-separated hex values for Design Studio swatches)
- & Upload hero banner images, set CTA text and links
- & Seed initial categories (T-Shirts, Hoodies, Mugs, Caps, Accessories, Gift Hampers)
- & Add first products with images (upload to R2 via admin Products panel)
- & Write first blog post (admin Blog !' New Post !' use AI Assist for first draft)
- & Add Testimonials (admin Testimonials !' Create)
- & Configure Telegram notification bot (Settings !' Telegram !' Test)

SEO Setup

- & Register https://trynexshop.com in Google Search Console
- & Submit /api/sitemap.xml URL to Google Search Console
- & Register in Bing Webmaster Tools; submit sitemap
- & Add Google Analytics ID in admin Settings !' Tracking
- & Add Meta Pixel ID in admin Settings !' Tracking
- & Verify OG image (1200×630px) loads at https://trynexshop.com/og-image.jpg
- & Check structured data with https://search.google.com/test/rich-results

§20 Replit Independence

Secret management
Cache

Replit Secrets panel
In-process Map
Replit Object Storage sidebar

Render env vars / GitHub Secrets
Upstash Redis, RES1
Cloudflare R2 (hard-exits if not set in prod)

Production Boot Guards

- Hard-exit if storage backend resolves to 'replit' in NOD
- Hard-exit if ALLOWED_ORIGINS is not set in producti
- Hard-exit if ADMIN_JWT_SECRET is < 32 chars in pr
- Hard-exit if any of DATABASE_URL, JWT_SECRET, A
- Warns (no exit) if GOOGLE_CLIENT_ID, FACEBOOK_

Migration to New Host

1. Export database: pg_dump \$DATABASE_URL > try
2. Set up new PG instance !' set DATABASE_URL to n
3. First server start will apply all 12 Drizzle migrations
4. Configure R2_* vars (or change to S3 / another S3-
5. Set all required env vars (see §17) — server will log
6. Run seed script: node scripts/seed.mjs (optional —
7. Update ALLOWED_ORIGINS and domain in Cloudf

